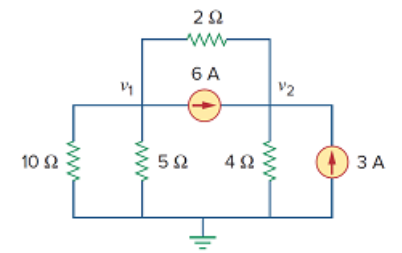


Problem

For the circuit in Fig. 3.51, obtain v_1 and v_2 .

Figure 3.51:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.50 in the textbook.

Apply Kirchhoff's voltage law at node, v_1 .

$$\frac{v_1}{10} + \frac{v_1}{5} + \frac{v_1 - v_2}{2} + 6 = 0$$

$$\frac{v_1 + 2v_1 + 5v_1 - 5v_2 + 60}{10} = 0$$

$$8v_1 - 5v_2 = -60 \dots\dots (1)$$

Apply Kirchhoff's voltage law at node, v_2 .

$$-3 - 6 + \frac{v_2 - v_1}{2} + \frac{v_2}{4} = 0$$

$$\frac{2v_2 - 2v_1 + v_2}{4} = 9$$

$$-2v_1 + 3v_2 = 36$$

$$v_2 = \frac{36 + 2v_1}{3} \dots\dots (2)$$

Step 2 of 3

Substitute $\frac{36 + 2v_1}{3}$ for v_2 in equation (1).

$$8v_1 - 5\left(\frac{36 + 2v_1}{3}\right) = -60$$

$$24v_1 - 180 - 10v_1 = -180$$

$$v_1 = 0\text{V}$$

Therefore, the node voltage, v_1 is 0V.

Step 3 of 3

Substitute 0 V for v_1 in equation (2).

$$v_2 = \frac{36 + 2(0)}{3}$$

$$v_2 = 12\text{ V}$$

Therefore, the node voltage, v_2 is 12 V.